

Katerina Smyrli

Robotics Engineer, PhD

✉ katerina@smyrli.com 📍 Athens, Greece 🔗 katerinasmyrli

Summary

Researching bipedal locomotion of robots and humans

Education

National Technical University of Athens , Control Systems Lab, Department of Mechanical Engineering	Athens, GR 2018 – 2025
Imperial College London , Tribology Lab, Department of Mechanical Engineering • Awarded with Distinction.	London, UK 2017 – 2018
National Technical University of Athens , Control Systems Lab, Department of Mechanical Engineering • Awarded with Distinction.	Athens, GR 2012 – 2017

Experience

Athena Research Center , Adjunct Researcher • Robotics Institute • HERON • The Hellenic Robotics Center of Excellence	Marousi, GR 2023 – present 3 years
Control Systems Lab, National Technical University of Athens , Robotics Researcher Researcher in multiple EU-funded as well as private research projects, focusing on Dynamics and Control in space robotics. • CoDyCO • Study of the Contact Dynamics during Object Capture on Orbit • Study of a Waste Bin Weighing System • Analysis and simulation of Waste Collector System • PRINCE • Passive Mechanical and Rendezvous Interface for Capture after End-of-Life • DiGi-CHE Asia • Enhancing Digital Capacities in Higher Education for Asian Universities	Athens, GR 2018 – present 8 years
TWI Hellas , Technical Project Manager Technical coordinator of Horizon Europe robotics projects. • PALPABLE • Multi-sensing tool for Minimally Invasive Surgery • SWAG • Soft Wearable Assistive Garments for Human Empowerment	Athens, GR 2023 – 2025 2 years
University of Duisburg-Essen , Visiting Researcher Part of a PhD exchange with a focus on clinical gait analysis (German Academic Exchange Service - DAAD). • Study of Gait Analysis Systems	Duisburg, DE 2018 – 2019 1 year

Publications

- Underactuated Virtual Gravity Control - Harnessing Passive Dynamics for Optimally Efficient Bipedal Locomotion** 2026
Aikaterini Smyrli, Evangelos Papadopoulos
- A Modular, Wireless and Wearable Biosignal Acquisition Platform** 2026
Antonios Doukakis, Aikaterini Smyrli, Makis Livadas, Henrique De Melo Ribeiro, Shadiya Alin-gal Meethal, Mohamad Reza Shahabian Alashti, Gabriella Lakatos, Patrick Holthaus, Farshid Amirabdollahian
- A Virtual Gravity Controller for Efficient Underactuated Biped Robots** 2025
Despoina Maligianni, Fotios Valouxis, Antonios Kantounias, Aikaterini Smyrli, Evangelos Papadopoulos
ieeexplore.ieee.org/document/11128793
- Low-Cost Markerless Gait Analysis Using a Minimal Human Model** 2025
Konstantina Tsintzira, Aikaterini Smyrli, Athanasios Mastrogeorgiou, Evangelos Papadopoulos
ieeexplore.ieee.org/document/11273621
- User-Centered Design Approach for Development of an Assistive Soft Exosuit and Initial Results on User Requirements** 2024
Gerdienke B. Prange-Lasonder, Kira Oberschmidt, Ilaria Pacifico, Marko Jamsek, Jan Babic, Federico Masiero, Lorenzo Masia, Aikaterini Smyrli, Erik C. Prinsen
link.springer.com/chapter/10.1007/978-3-031-77588-8_13
- Guidelines For Optimal Human Mesh Generation Using Deep Learning-Driven Avatar Reconstruction For Gait Analysis** 2024
Odysseas Stavrakakis, Athanasios Mastrogeorgiou, Aikaterini Smyrli, Evangelos Papadopoulos
ieeexplore.ieee.org/document/10769832
- Gait Analysis with Trinocular Computer Vision Using Deep Learning** 2023
Odysseas Stavrakakis, Athanasios Mastrogeorgiou, Aikaterini Smyrli, Evangelos Papadopoulos
ieeexplore.ieee.org/document/10328370
- Modeling, Validation, and Design Investigation of a Passive Biped Walker with Knees and Biomimetic Feet** 2022
Aikaterini Smyrli, Evangelos Papadopoulos
ieeexplore.ieee.org/document/9812129
- Improved biped walking performance around the kinematic singularities of biomimetic four-bar knees** 2022
Aikaterini Smyrli, Evangelos Papadopoulos
ieeexplore.ieee.org/document/9982027
- Development of a passive biped robot digital twin using analysis, experiments, and a multibody simulation environment** 2021
Christos Vasileiou, Aikaterini Smyrli, Anargyros Drogosis, Evangelos Papadopoulos
www.sciencedirect.com/science/article/abs/pii/S0094114X2100104X

- A Generalized Model for Compliant Passive Bipedal Walking - Sensitivity Analysis and Implications on Bionic Leg Design** 2021
Aikaterini Smyrli, Georgios A. Bertos, Evangelos Papadopoulos
asmedigitalcollection.asme.org/biomechanical/article-abstract/143/10/101008/1109461/A-Generalized-Model-for-Compliant-Passive-Bipedal
- A methodology for the incorporation of arbitrarily-shaped feet in passive bipedal walking dynamics** 2020
Aikaterini Smyrli, Evangelos Papadopoulos
ieeexplore.ieee.org/document/9196617
- On the effect of semielliptical foot shape on the energetic efficiency of passive bipedal gait** 2019
Aikaterini Smyrli, Mehdi Ghiassi, Andres Kecskemethy, Evangelos Papadopoulos
ieeexplore.ieee.org/document/8967565
- Application of OpenPose deep learning algorithm for gait parameter identification** 2019
Athanasios Mastrogeorgiou, Aikaterini Smyrli, Evangelos Papadopoulos, Andres Kecskemethy, Silke Gegenbauer
nereus.mech.ntua.gr/pdf_ps/eccomas19.pdf
- Efficient Stabilization of Zero-Slope Walking for Bipedal Robots Following Their Passive Fixed-Point Trajectories** 2018
Aikaterini Smyrli, Georgios A. Bertos, Evangelos Papadopoulos
ieeexplore.ieee.org/document/8460845

Theses

- Design and control of anthropomorphic and energy-efficient robotic systems with applications in lower-limb prostheses.** 2025
The thesis investigates bipedal locomotion through modeling, simulation, and control of bio-inspired walkers for applications in robotics and lower-limb prosthetics. It develops bio-mimetic foot and knee designs, informed by human gait data, and an energy-efficient control strategy to create robotic and prosthetic systems that effortlessly generate natural walking patterns.
Aikaterini Smyrli
phdtheses.ekt.gr/eadd/handle/10442/60169?locale=en
- Robotic Grip - Design, development and evaluation of a tactile sensing device.** 2018
This thesis investigates the design and development of a finger-like tactile sensing device for robotic manipulators to detect contact and predict slip. Integrating thermal and force sensors inspired by the human tactile system, the device is experimentally evaluated for contact detection and slip sensing across different forces and materials.
Aikaterini Smyrli
spiral.imperial.ac.uk/entities/publication/fa9e6e52-efb4-44c9-912f-6cd14535f826
- Design, modeling and control of a bipedal walking robot with applications in human walking dynamics.** 2017
This thesis investigates the passive dynamics of a bipedal walking model with compliant legs, semicircular feet, and a hip mass to better replicate human gait, particularly the double-stance phase and heelstrike dynamics.
Aikaterini Smyrli
dspace.lib.ntua.gr/xmlui/handle/123456789/46119?locale-attribute=en

Skills

Robotics

Workshop Organizer

Tailored to Move - Wearable Robotics for Motion Assistance	2026
Advanced tissue examination for diagnostics in surgical soft robotics	2025
Markerless Motion Analysis	2023

Awards

C. Mavroidis Award For excellence in Robotics and Automation. C. Mavroidis	2022, 2020
D. Thomaidis Award For the publication of several research works. National Technical University of Athens	2018-2022
Thomaidis Foundation Award For excellent academic performance. Thomaidis Foundation	2017
C. Papakyriakopoulos Award For excellence in Mathematics. National Technical University of Athens	2013
Scholarship from the Association of Greeks from Egypt For excellent academic performance. Association of Greeks from Egypt	2012-2016

Languages

Greek
Native speaker

English
Proficient (C2)

Italian
Basic (B2)

Interests

Crafts

Activities

Projects

Quantum Computing Quantum computing is the use of quantum-mechanical phenomena such as superposition and entanglement to perform computation. Computers that perform quantum computations are known as quantum computers. • Quantum Teleportation • Quantum Cryptography	Jan 2018 – Jan 2018
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References

Professor John Doe

Professor Jane Smith